

1. COMPLETE MODEL STACK FOR VIVE

The model stack separates speech gating, voice authenticity, speaker consistency, speech recognition, semantic understanding, uncertainty and risk fusion. Large pretrained models are used where appropriate rather than training foundation models from scratch.

Model/layer	Purpose	Input → output	Model type	Role in VIVE
Silero VAD	Speech / non-speech	Audio → speech segments	Pretrained	Continuous speech gating and inference reduction.
AASIST	Synthetic/manipulated speech	Speech waveform/features → anti-spoof evidence	Pretrained first; evaluate/adapt as required	Primary acoustic voice-authenticity evidence.
ECAPA-TDNN	Speaker consistency	Speech → speaker embedding → similarity to reference	Pretrained	Identity consistency only when an authorized reference exists.
IndicConformer 600M	Multilingual ASR	Speech → transcript	Pretrained	Hindi/Tamil/Indic speech-to-text; language-aware semantic input.
Intent classifier	Caller objective	Transcript/context → intent probabilities	Project-trained classifier	Detect OTP, password, card, banking credential, transfer, account change, remote access, urgent action, threat, confidential information, normal and unknown.
Behavior classifier	Social-engineering behavior	Transcript/context → behavior probabilities	Project-trained classifier	Detect authority impersonation, urgency, threat, fear, secrecy, pressure, reward promise and normal behavior.
OOD / uncertainty layer	Unknown/insufficient evidence	Scores/embeddings → state	Project layer	Handle unfamiliar patterns and insufficient evidence without forced classification.
Risk fusion	Evidence → risk/confidence	Evidence vector → risk + confidence + reasons	Transparent fusion; calibrated/learned fusion planned	Combines synthetic, speaker, intent, behavior, context, temporal, uncertainty and audio-quality evidence.
Temporal risk	Persistence/escalation	Packet risk history → temporal risk	Project logic	Rolling/EMA behavior, hysteresis, corroboration and escalation timing.

2. MODEL SOURCES / REFERENCES

Model	Primary source / reference	Use
Silero VAD	Silero VAD pretrained model	Speech segmentation
AASIST	Clova AI / SpeechAntiSpoofingBenchmarks AASIST	Anti-spoof evidence
ECAPA-TDNN	SpeechBrain spkrec-ecapa-voxceleb	Speaker embeddings
IndicConformer	AI4Bharat indic-conformer-600m-multilingual	Indic ASR
Intent / Behavior	Project-trained classifier built from the project semantic dataset strategy	Social-engineering semantics
Risk/OOD/Temporal	VIVE project implementation	Decision-support and uncertainty layer

3. DATA SOURCES PLANNED FOR THE PROJECT

Dataset/resource	Primary purpose	Planned VIVE use
ASVspoof releases / ASVspoof 5	Synthetic/spoof speech	Anti-spoof training/evaluation and generator-disjoint testing.
IndicSynth	Indian synthetic speech	Indic synthetic-voice evaluation/adaptation.
Vaani	Indian speech resources	Genuine Indian-language speech evaluation/training where access/license permits.
VoxCeleb1/2	Speaker speech	Speaker embedding/reference evaluation and speaker foundation.
FLEURS	Hindi/Tamil speech	ASR component testing; Hindi hi_in and Tamil ta_in.
LibriSpeech	Clean English speech	Clean ASR/speaker testing.
ScamShield	English/Hindi/romanized Hinglish scam text	Intent/behavior classifier data after license/provenance review.
Hinglish/English Financial Scam/Fraud Text Dataset	Financial/KYC/UPI/OTP/fake-loan/fake-job/emergency scam text	Intent/behavior classifier data after label mapping and leakage review.
Consented/project-generated samples	Project-specific speech/scenarios	Realistic evaluation, human-scam and synthetic-legitimate testing.